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SVD as a Decomposition
SVD as a Data Analytics Tool
Assignment 5
Quiz: Singular Value Decomposition

Singular Value Decomposition

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Due Jan 28, 11:59 PM IST **Attempts** 3 every 8 hours

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1. What is the direction of maximum dilation (direction of largest growth, relative to the input) for **1 point**

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

- ○ direction: $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
- ○ direction: $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
- ◉ direction: $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
- ○ direction: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

2. Which picture best represents the image of the unit circle under the mapping circle -> A * circle, where **1 point**

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

- ○ Horizontal ellipse
- ◉ Vertical ellipse
- ○ Ellipse at 45 degree angle (plus or minus)
- ○ Circle
- ○ Ellipse at correct angle

3. **1 point**

For a rotation matrix $A = \begin{bmatrix} 0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix}$, what is the image of the unit circle:

- ○ Ellipse
- ◉ Circle
- ○ None of the above

4. Q: **1 point**

Construct the rotation matrix R that rotates by $30°$ in the **counter-clockwise** direction.

- ○ R = $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 1/2 & -\sqrt{3}/2 \end{bmatrix}$
- ○ R = $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$
- ◉ R = $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$
- ○ R = $\begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$

5. Q: **1 point**

Given the SVD

$$A = U \cdot \Sigma \cdot V^T = \begin{bmatrix} \sqrt{1/2} & \sqrt{1/2} \\ -\sqrt{1/2} & \sqrt{1/2} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2/9} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix},$$

where $\sqrt{1/2} = 0.707107\ldots$, and $\sqrt{3}/2 = 0.866025\ldots$.
Answer the following by reading it off this SVD (no arithmetic required)

What is the direction of maximum dilation?

- ○ direction = $\begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}$
- ◉ direction = $\begin{bmatrix} \sqrt{3}/2 \\ 1/2 \end{bmatrix}$
- ○ direction = $\begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$
- ○ direction = $\begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$

6. Q: **1 point**

Given the SVD

$$A = U \cdot \Sigma \cdot V^T = \begin{bmatrix} \sqrt{1/2} & \sqrt{1/2} \\ -\sqrt{1/2} & \sqrt{1/2} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2/9} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix},$$

where $\sqrt{1/2} = 0.707107\ldots$, and $\sqrt{3}/2 = 0.866025\ldots$.
Answer the following by reading it off this SVD (no arithmetic required)

What is the maximum amount of dilation possible (i.e., could the length of $A\mathbf{v}$ be twice the length of $\mathbf{v}$? if not, how much bigger could $A\mathbf{v}$ relative to the length of the input $\mathbf{v}$?) (Hint: look at the entries in Σ.)

- ◉ answer = sqrt(2) = 1.41421....
- ○ answer = 2.0
- ○ answer = $\sqrt{2 + 2/9}$
- ○ answer = $\sqrt{3}/2$

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